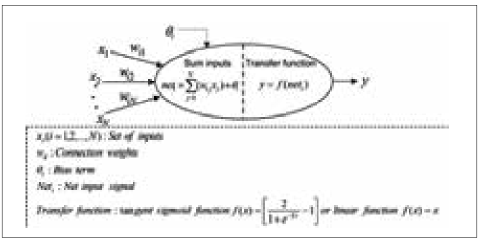
The success of a neural network is determined with how quickly it generates these weights, how well it updates them and how well it adapts to new data.

Jargon and terminologies

It's common for a person to lose track of the scientific and analytic words used in the neural networking atmosphere. Here are a few definitions to get you started:

Perceptron: A perceptron is a neuron that accepts only binary inputs but produces a single binary output.

Learning Rate: The learning rate determines the speed with which weights and biases are updated with each iteration. You might be tempted to increase the learning rate when creating a neural network but run the risk of the derivative missing the 0 slope point.



Parameters and Outputs In a Neuron

Rectified Linear Unit: An activation that outputs 0 for negative values and a linear equation for positive input values. It's considered to be more effective than tangential and sigmoidal neurons.

Mean Squared Error: Sum of the squared errors of the expected outputs from the actual outputs divided by the number of training inputs across the network.

Weights and Biases: Numbers and constants that are multiplied to each input are the weights. Biases are the constants added that later become the activation function. The activation function can use various types of algorithms among the most common which include- gradient descent, scalar gradient descent, linear propagation, conjugate gradient, tangential hypotenuse and so on.

Affine Layer: A fully-connected layer where every input node is connected with every other input node.

Autoencoder: An autoencoder predicts the input backwards to introduce a bottleneck in the system. They play important roles in PCA(principal component analysis).